

requiring a countdown timer. I'll demonstrate how to do both of these using the Slider Puzzle game pictured in [Figure 13-3](#).



*Figure 13-3: The Slider Puzzle user interface*

When you start this game, the tiles are randomly rearranged, and there is one empty black space. The goal of the game is to move tiles one at a time to put them in order from 1 to 15. You are only allowed to click a tile that is horizontally or vertically adjacent to the empty square. Clicking a valid tile swaps it with the space. I won't get into the details of the full implementation of the game (although the source code is available online, with the rest of the book's resources). Instead, I will focus on how to integrate a timer.

The `pyghelpers` package contains two classes that allow programmers to track time. The first is `CountUpTimer`, which starts at zero and counts up indefinitely, or until you tell it to stop. The second is `CountDownTimer`, which starts at a given amount of time and counts down to zero. I've built a version of the game for each. The first version lets the user see how long it takes them to solve the puzzle. In the second one, the user is given a certain amount of time when they start the game, and if they have not completed it when the timer reaches zero, they lose the game.